

1. Network functional areas

Access Network

The **access network** is the part of the communication network that connects the subscriber to the immediate service provider.

It is the network portion closest to the end user.

It may be divided into:

- **feeder/distribution network**
- **drop/edge network**

Typical access technologies are:

- copper / DSL
- fiber
- wireless
- cellular
- satellite
- cable

The main role of the access network is therefore:

user → service provider connection

The slides explicitly distinguish it from the core network.

Core Network

The **core network** is the backbone of the network.

Typical characteristics:

- usually **mesh topology**
- provides **any-to-any connectivity**
- contains switches or IP routers
- usually relies on an **optical backbone**

The Internet can be seen as an interconnection of many core networks.

Edge

The **edge** is located between the access part and the core.

It can implement **intelligent functions** that are not performed inside the core.

Example: in an MPLS network, an edge device can examine packets and select the path they should follow through the core.

Think:

Access → Edge → Core

The edge can be relatively **smart**, while the core can perform simpler high-speed switching operations.

QUIZ FACT

EDGE

What is the EDGE of a network and which is its role?

☐

That part of the network in the between of the access and the core with some intelligent functions

☐

The very peripheral part of the network where all the applications are

☐

The very peripheral part of the network where only simple functions are implemented

☐

The very core part of a network where powerful network functions are implemented

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2. Access technologies

You do not need to memorize huge descriptions. Remember the association.

Technology	Medium / Key point
Wired	physical cables
Wireless	radio signals
Satellite	satellites
Fiber optic	light through fiber
DSL	existing telephone copper lines
Cable	coaxial cable
Cellular	mobile radio network
Powerline	electrical wiring
5G	high-speed, low-latency cellular access

Main trade-offs

Wired

- reliable
- stable
- high speed

Wireless

- mobility
- flexibility
- easier deployment

Fiber

- high bandwidth
- low latency
- stable transmission

DSL

- uses existing telephone infrastructure
- widespread
- cheaper than deploying completely new infrastructure

Satellite

- very wide geographical coverage
 - useful in remote areas
-

3. Evolution of fixed access networks

Historically, access networks were mainly based on **copper pairs** used for telephone service.

Today the main evolution is: **Copper → Fiber**

The reason is that fiber offers:

- higher capacity
- higher speed
- stable transmission
- shorter or zero copper segment

QUIZ FACT

The evolution of the current access network aims at:

☐

Employing fiber cables instead of copper ones

☐

Using only wireless access given its flexibility and low cost

☐

Using in a more capillary way the already existing infrastructure based on copper cables

☐

Providing an ADSL access to all the users

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4. FTTx

FTTx = Fiber To The x

The letter tells you **how close fiber gets to the user**.

Important versions:

Name	Meaning
FTTE	Fiber To The Exchange
FTTC	Fiber To The Curb
FTTB	Fiber To The Building
FTTH	Fiber To The Home
FTTP	Fiber To The Premises
FTTN	Fiber To The Node / Neighborhood

The fundamental rule is:

The closer fiber gets to the user, the shorter the copper segment and the higher the achievable bit rate.

The slides show, for example, that FTTE leaves roughly 1.2–1.5 km of copper, while FTTCurb reduces it to tens of metres.

Expected bit rate

Among:

- ADSL
- FTTE
- FTTC
- FTTB

the best expected bit rate is **FTTB**, because fiber reaches the building and very little copper remains.

TEST FACT

FTTX

Which one of these configurations is the best in terms of expected bit rate?

☐

FTTB - to the building

☐

FTTE- to the exchange

☐

FTTC -to the curb

☐

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Deployment cost

Same idea, reversed from a cost perspective.

More fiber deployed closer to users → generally **higher deployment cost**.

Among FTTE, FTTC and FTTB:

FTTB has the highest deployment cost in the provided test.

So remember:

more fiber → more cost to deploy → better performance

5. Optical access: OLT, ONU, ONT, ODN

Important acronyms:

- **OLT — Optical Line Terminal** → Located on the operator/network side.
- **ONU — Optical Network Unit** → Terminates or converts the optical access closer to users.
- **ONT — Optical Network Termination** → Optical termination at the user side.
- **ODN — Optical Distribution Network** → The optical distribution infrastructure between OLT and users.

These are the main elements shown in the FTTx reference architecture.

6. AON vs PON

AON — Active Optical Network

Also presented as: **Point-to-Point (P2P)**

Contains active equipment in the distribution network.

PON — Passive Optical Network

Uses:

- passive optical splitters
- tree-based topology
- no powered active switching elements inside the optical distribution path

The slide definition is essentially: **passive branching of fibers through optical splitters.**

Main advantage of PON

Past quiz answer: **lower infrastructure cost**, because fewer active components are required.

Do NOT think:

PON means there is no ONU.
ONU/ONT devices are still needed.

Multiple ONUs transmitting

A previous quiz asks what happens if several ONUs want to transmit simultaneously.

Correct idea: **The OLT schedules ONU transmissions to avoid collisions.**

So do not think of Ethernet-style random collisions and retransmissions as the normal PON solution.

7. Wavelength Division Multiplexing — WDM

Basic idea: **Different wavelengths of light can coexist in the same optical fiber.**

Think of them as different **colors**.

If a fiber supports several wavelengths, multiple optical channels can be transmitted simultaneously over that same physical fiber.

QUIZ SBERS

How the Wavelength Division Multiplexing (WDM) is a fiber-optic transmission for a PON is used?

Select one:

- ☐ a. If we have 10 different wavelenghts we can have 10 different colors contemporalry in the same fiber.
- ☐ b. The same Optical Network Unit can trasmit different pakets on different wavelenghts.
- ☐ c. One wavelenght is used for the downstream and one for the upstream.
- ☐ d. If we have 10 different wavelenghts we can have 10 different fibers contemporalry in the same cable.

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8. DSL / xDSL

DSL = Digital Subscriber Line

It is a family of technologies that transmit digital data using the existing wires of the telephone access network.

Important family members:

- HDSL
- SDSL
- ADSL
- ADSL2+
- VDSL
- VDSL2

The main differences are:

- bit rate
- distance
- symmetry/asymmetry
- frequency range
- equipment location

Why ADSL became successful

This is a very likely question because it appears repeatedly in the provided quizzes.

The main reason: **ADSL reuses the existing copper telephone infrastructure.** It did NOT require replacing all copper cables with fiber.

QUIZ FACT

ADSL success

One of the key element of the ADSL success is related to:

☐

The use of the same copper cable of the old telephone network

☐

The use of the same frequency band of the old telephone network

☐

The use of new fiber cables in place of the old telephone ones

☐

The fact that no new network elements are needed at the Central Office side

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Important trap:

ADSL uses the same **cable**, but NOT simply the same frequency band as telephone voice.

Why ADSL is asymmetric

A = Asymmetric

The downstream data rate is greater than the upstream data rate.

Why?

Traditional residential Internet traffic requires more data in **network → user** rather than **user → network**

Examples:

- browsing
- video
- downloads

The slides explain that the distinguishing feature of ADSL is exactly the different amount of data flow in the two directions.

ADSL frequency bands

Telephone voice and ADSL data coexist on the same copper pair because they use **different frequency bands**.

POTS

Traditional voice occupies roughly: **0–4 kHz**

ADSL

UPSTREAM: approximately **25.875–138 kHz**

DOWNSTREAM: Approximately **138–1104 kHz**

So:

- **low frequencies → voice**
 - **higher frequencies → data**
 - **upstream and downstream → separated bands**
-

9. Splitter

A **POTS splitter** separates:

- telephone voice
- DSL data

It uses filtering so that telephone devices receive the low-frequency voice band while the DSL modem receives the higher-frequency data band.

Its main purpose is: **prevent interference between POTS and DSL**.

The slides describe both a dedicated splitter and distributed microfilters.

10. Cross-talk

This is one of the most important quiz topics.

Copper pairs are grouped together inside cables.

Signals travelling on one pair can electromagnetically couple into another pair.

Therefore:

Cross-talk = undesired interference caused by coupling between signals on different circuits/pairs.

QUIZ FACT

Cross_talk

What is the cross-talk noise on ADSL?



The coupling of signals from a circuit to another giving rise to undesired interference



A noise that is generated at a generic receiver within a fiber cable



A noise that is generated at a generic receiver within a copper cable



The interference of a signal in a frequency band to signals in other frequency bands of the ADSL

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Factors affecting cross-talk

It depends on:

- transmitted power spectral density
- number of twisted pairs
- overlapping frequency bands

And: **cross-talk generally becomes more important at higher frequencies.**

That is why it becomes particularly problematic for high-speed DSL technologies.

11. NEXT vs FEXT

There are two main forms.

NEXT — Near-End Cross-Talk

Transmitter and affected receiver are on the **same side** of the cable.

It can be particularly strong because the disturbing transmitter is physically close to the receiver.

NEXT is one reason ADSL separates upstream and downstream frequencies.

Shortcut: **N = Near = same side**

FEXT — Far-End Cross-Talk

Transmitter and affected receiver are on **opposite sides**.

The interfering signal travels through the cable before affecting the receiver.

Shortcut: **F = Far = opposite sides**

Very useful memory rule

- **NEXT → TX and RX same end**
 - **FEXT → TX and RX opposite ends**
-

12. DMT — Discrete Multi-Tone

DMT is the main ADSL modulation concept you need to know.

Instead of using one large frequency channel, DMT divides the available bandwidth into many small **subchannels**.

Each subchannel has a carrier called a: **tone**

Data are transmitted independently on these tones using QAM.

The slides describe ADSL DMT as approximately:

- **256 sub-bands**
- about **4.3125 kHz each**

Main advantage

Each subchannel can be adapted independently according to channel quality.

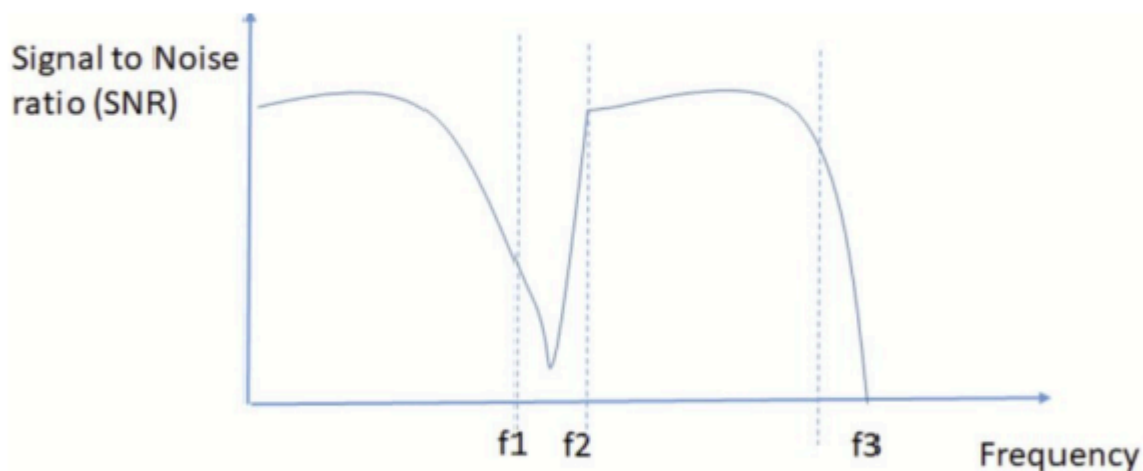
If one frequency range suffers from:

- noise
 - interference
 - strong attenuation
- that tone can receive fewer bits or be completely unused.
- Good channel: **higher-order QAM → more bits**
 - Poor channel: **lower-order modulation / fewer bits**

- Very poor channel: **do not use the tone**

Question 4: ADSL channel behavior and Discrete Multitone Technology (DMT)

Considering an ADSL line with frequencies fully attenuated (low SNR) in certain ranges, how should a modulation based on Discrete Multitone Technology (DMT) handle these frequencies?



- a) It should include transmitting "tones" in the frequency band below f_1 and between f_2 and f_3 .
- b) It must include only transmitting "tones" in the frequency band between f_2 and f_3 .
- c) It should include transmitting "tones" only in the frequency band below f_1 .

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13. Water Filling

Water filling determines how transmission power should be allocated among the different DMT subcarriers.

The intuition: **allocate more useful resources where the channel is better and less where the channel is bad.**

It is particularly useful when channel quality is **not flat across frequencies.**

QUIZ FACT

Question 7: Water filling algorithm in ADSL

When are the advantages of applying the water filling algorithm in an ADSL system most noticeable?

- a) The channel behavior both in the upstream and downstream has a flat (constant) signal-to-noise ratio.
- b) It is important to separate the telephone bandwidth from the one used by the data.
- c) It is important to split the transmission into two parts: downstream and upstream.
- d) The channel behavior both in the upstream and downstream has not a flat (constant) signal-to-noise ratio.

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D

If all frequencies behaved identically, there would be much less reason to perform adaptive allocation.

14. ADSL architecture

Important components:

ATU-R

ADSL Transmission Unit — Remote

User side.

Essentially the user's DSL modem/termination.

ATU-C

ADSL Transmission Unit — Central Office

Operator side.

DSLAM

DSL Access Multiplexer

Located in the operator network.

Main roles:

- contains many ATU-C interfaces

- multiplexes users' traffic
- demultiplexes traffic
- negotiates line speed
- provides management functions

The slide reference model explicitly connects the ATU-R, copper line, splitter, ATU-C and DSLAM.

15. Distance vs DSL bit rate

This relation is fundamental:

longer copper line → more attenuation → lower achievable bit rate

Therefore:

shorter copper → higher speed

This explains many topics at once:

- why fiber is brought closer to the user
- why FTTC is better than FTTE
- why FTTB is better than FTTC
- why VDSL requires short copper loops

For VDSL, the slides explicitly say:

| the key to the fastest bit rate is the shortest possible copper length.

16. VDSL

VDSL = Very-high-speed Digital Subscriber Line

It uses a much larger frequency range than ADSL and can provide much higher bit rates.

But there is a cost: **higher frequencies suffer more attenuation and cross-talk.**

Therefore VDSL works best when copper is short.

Typical architecture: **fiber → cabinet/ONU/DSLAM → short copper → user**

rather than: Central Office → long copper → user

17. ADSL vs ADSL2+ vs VDSL vs VDSL2

The slide gives these approximate values:

Technology	Frequency	Maximum rates
ADSL	up to ~1.1 MHz	~8 Mbps down
ADSL2+	up to ~2.2 MHz	~24 Mbps down
VDSL	up to ~12 MHz	~55 Mbps down
VDSL2	up to ~30 MHz	~100 Mbps down/up

The conceptual relation is much more important than the exact number:

ADSL < ADSL2+ < VDSL < VDSL2

means generally: **larger bandwidth → potentially higher bit rate**

but also: **higher frequencies → shorter useful copper distance / greater cross-talk problems.**

18. VDSL vectoring

Another very likely topic.

VDSL uses high frequencies, so **cross-talk becomes a major limiting factor.**

Vectoring attempts to cancel cross-talk.

It works by generating an appropriate **anti-signal** for each cross-talk-impaired line.

Requirements include:

- synchronization
- knowledge of the different lines
- estimation of cross-talk coefficients
- calculation of cancellation signals

The slides explicitly describe vectoring as cross-talk cancellation through an anti-signal.

QUIZ SBURS

Which are the two improvements enabling the VDSL?



The vectoring of the signals in the copper cables and the placement of the DSLAM close to the end user



The vectoring of the signals in the fibers cables that arrive till the end users



The use of high frequencies in the copper cable band and the full cancellation of the cross-talk noise



The use of high frequencies in the fiber optic cables and the vectoring of these signals

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RICORDA

1. **Vectoring on copper lines**
2. **DSLAM/fiber termination closer to the end user**

19. Wireless access — concepts likely to be tested

From the provided past tests, the professor prefers conceptual questions rather than formulas here.

Interference

When several users share a wireless system, the critical issue is often:
interference between users if multiple access is not properly controlled.

This is the answer given in the past quiz.

Do not confuse:

- **path loss** = signal gets weaker with distance
- **multipath** = multiple delayed copies of the same signal
- **interference** = unwanted signals from other transmissions

Multipath

A wireless signal may reach the receiver through several propagation paths because of:

- reflection
- diffraction
- scattering

Different paths have different:

- attenuation
- delay

From the supplied quiz, if asked what must be modelled to use multipath constructively:
attenuation and delay of the different paths.

20. Latency

Latency measures how long the network takes to react/deliver information.

It is different from throughput.

Throughput

How much data can be transferred per unit of time.

Latency

How quickly communication can happen.

Future applications increasingly require very fast interaction with the network.

Therefore latency is important because many applications require **high responsiveness.**

Very common trap:

| high throughput

does NOT mean:

| low latency

They are different performance metrics.

21. LEO vs GEO satellites

This is in the current slide set, so I would know the basic differences.

LEO — Low Earth Orbit

Altitude: ~500–2000 km

Latency: ~20–40 ms

Advantages:

- low latency
- high bandwidth potential
- global coverage possible

Problems:

- many satellites required
- shorter lifespan
- deployment/maintenance cost
- space debris

GEO — Geostationary Earth Orbit

Altitude: ~35,786 km

Latency: ~500+ ms

Advantages:

- huge coverage
- one satellite can cover about one third of Earth
- longer lifetime

Mental shortcut

LEO = Low altitude = Low latency

GEO = far away = high latency

22. PPP — lower priority, but know the basics

The last part of 02 is labelled as an appendix, so I would study this after the main topics.

PPP = Point-to-Point Protocol

Used to transport packets between two peers.

In ADSL it can support functions such as:

- authentication
- authorization
- network configuration

Two common forms mentioned:

- **PPPoA**
 - PPP over ATM
- **PPPoE**
 - PPP over Ethernet

LCP

Link Control Protocol

Handles:

- establishment
- configuration
- control
- termination of the PPP link

NCP

Network Control Protocol

Configures network-layer protocols.

PAP vs CHAP

PAP

- username/password style
- less secure

CHAP

- challenge-response
- hashing
- more secure than PAP

23. QAM — minimum you need

QAM = Quadrature Amplitude Modulation

More constellation points → more bits per symbol.

Example from the slide: **16-QAM → 16 symbols → 4 bits/symbol**

because: $\log_2(16) = 4$

In DMT:

- good subchannel → larger QAM constellation
- poor subchannel → smaller constellation

High-yield quiz traps

These are the distinctions I would have very clear before Friday:

If the question says...	Think...
ADSL success	reuse existing copper cable
Cross-talk	coupling between different pairs
NEXT	TX/RX on same side
FEXT	TX/RX on opposite sides
DMT	many independent tones/subchannels
Bad DMT frequency	reduce/disable that tone
Water filling	useful with non-flat SNR
VDSL high speed	short copper
VDSL vectoring	cross-talk cancellation
FTTx higher bit rate	fiber closer to user
FTTx higher deployment cost	fiber closer to user
PON	passive splitters
PON multiple ONUs	OLT scheduling
WDM	several wavelengths/colors in one fiber
EDGE	intelligent part between access and core
Latency	responsiveness
Wireless multiple users	interference
LEO	low altitude, low latency
GEO	high altitude, high latency
DSLAM	aggregates/multiplexes DSL lines

If the question says...	Think...
Splitter	separates voice and DSL frequencies